

GATI SHAKTI VISHWAVIDYALAYA

C Programming Language

Lab Assignment – 8

Instructions:

- 1. Write all the program and execute.*
 - 2. Maintain soft copy of all codes and prepare a Lab Record consisting all hand written programs.*
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1. Write a program to store the details of 5 students. Use nested structure for storing the address. Print all the details of the students whose roll number the user enters.
2. Create a structure called point to represent a point in 2D space with 'x' and 'y' coordinates. Write functions to calculate the distance between two points and the midpoint of a line segment given by two points.
3. Declare a structure book to store details of a book (title, author, price). Write a program to create an array of 'n' books, where 'n' is provided by the user. Prompt the user to enter details for each book and display them back.
4. Define a structure rectangle to represent a rectangle with length and width. Write a program to create a structure pointer, allocate memory dynamically, initialize its members, and calculate the area of the rectangle.
5. Define a structure time to represent time (hours, minutes, seconds). Write functions to add two time values and display the result. Use pass-by-reference to modify the original time structures.
6. Define two structures: date to store a date (day, month, year) and employee to store information about an employee (name, ID, date of joining). Write a program to create a structure variable of type employee and initialize its members, including the date of joining.
7. Write a program using file handling to check whether a file already exist or not.
8. Write a program to read an already existing file using various method fscanf(), fgetc(), fgets() and fread().
9. Write a program to write in a file using various method fprintf(), fputc(), fputs() and fwrite().
10. Write a program to print the last char of an already existing file using file handling.